Improving Fine-Grained Food Classification via Hybrid Label Smoothing and EMA Distillation in Progressive CNN Fine-Tuning

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Abstract

Fine-grained food classification remains a challenging task in computer vision due to high intra-class variance and subtle inter-class differences. In this paper, we propose a novel training framework designed to enhance the performance of lightweight convolutional neural networks for food recognition tasks, particularly focusing on the EfficientNet-B0 model. Our proposed method integrates **hybrid label smoothing** with **exponential moving average (EMA) teacher-student distillation** in a **progressive fine-tuning** strategy. Evaluations conducted on the UEC-Food100 dataset demonstrate consistent improvements in both Top-1 and Top-5 accuracy compared to standard baselines. The results highlight the effectiveness of the approach in achieving a favorable trade-off between recognition accuracy and model efficiency, making it well-suited for deployment in resource-constrained environments.

1 Introduction

Food has become an indispensable part of human life and also the foundation of distinctive culinary cultures worldwide. Cuisine not only embodies cultural beauty but also serves as a universal language that connects people across borders. As individuals whose enthusiasm for gastronomic cultures, we are fascinated with Japanese cuisine - one of the most diverse culinary traditions in the world. Motivated by the desire to share knowledge about Japanese food culture, we intend to develop a mobile app that helps users to identify Japanese dished from self-captured photos. The core challenge of this project lies in building a highly accurate fine-grained food classification system capable of predicting labels that closely match real-world dishes.

Fine-grained food classification refers to the task of assigning precise and specific food labels to input images. This task has received increasing attention from many researchers in the computer vision community due to its practical applications in dietary assessment, smart restaurants, and food logging. However, it remains a highly challenging task due to the inherent visual complexity of food images. Many dishes from different categories may appear extremely similar in colors, presentation, or ingredients - that's called inter-class similarity. Besides, something unexpected known as intra-class variance may happen when images of dishes within the same category can vary widely due to external factors such as lighting conditions, camera angle, portion size, or even cooking style. All these phenomena significantly degrade the performance of conventional classification models, especially when applied to real-world scenarios.

Recent advances in convolutional neural networks (CNNs), particularly the EfficientNet family, have demonstrated strong potential in balancing model accuracy and computational efficiency. However, training such models effectively for fine-grained food classification remains a significant challenge. Despite leveraging modern architectures, CNNs are still prone to overfitting—especially in the

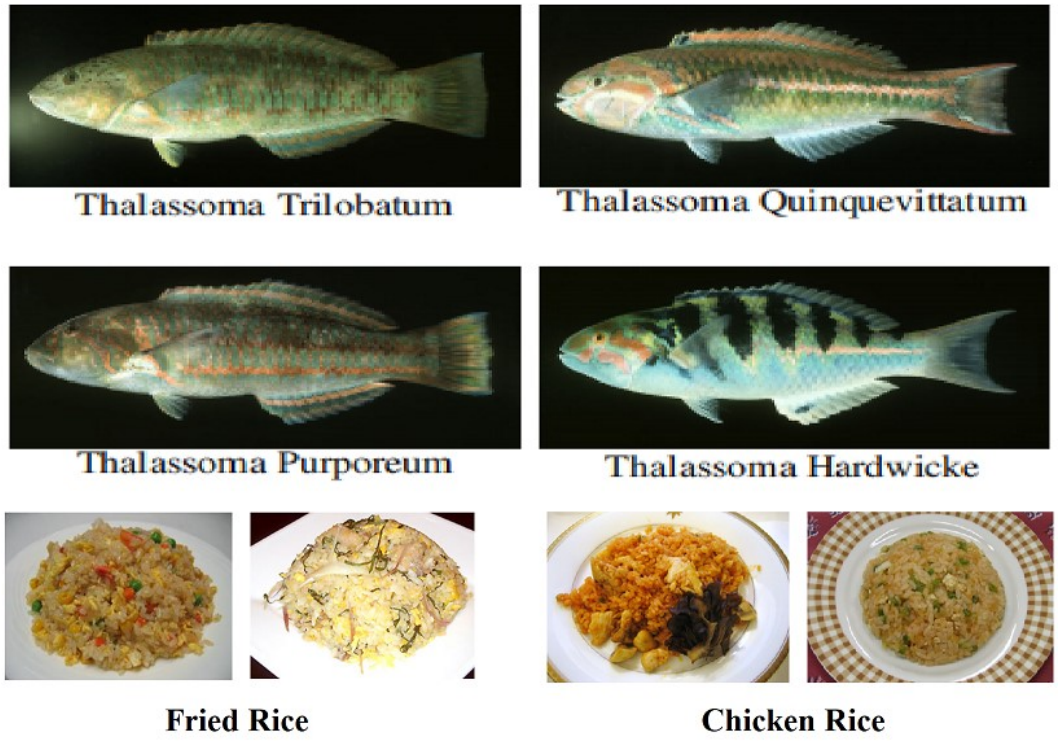


Figure 1: The top four images show example images of four fish species whose low inter-class variation; the rest images show examples for two kinds of dishes, with each dish type having high intra-class variation.

presence of limited data or class imbalance — and require careful optimization to achieve good generalization.

Selecting the most appropriate dataset also plays a crucial role in the success of this experiment. Since our task focuses on Japanese food classification, it is essential to use a dataset that contains diverse and representative examples of Japanese cuisine. After considering various options, we chose the UEC-Food100 dataset, which is one of the most widely used benchmarks for fine-grained Japanese food classification. This dataset contains over 14,000 images across 100 distinct categories, covering a wide variety of Japanese dishes such as sushi, ramen, donburi, and regional specialties. Each image is annotated with both class labels and bounding boxes, allowing us to utilize cropped regions for more focused and accurate training. Besides, UEC-Food100 has been extensively adopted in many previous works on food classification. Using the same dataset enables us to make direct and fair comparisons with existing methods, thereby providing a clearer evaluation of our approach's effectiveness.

In this paper, we present a lightweight and effective training framework tailored for fine-grained food classification. Specifically, we apply a progressive fine-tuning strategy to a pre-trained EfficientNet-B0—the most compact variant in the EfficientNet family(1). In addition, we incorporate hybrid label smoothing and exponential moving average (EMA) distillation to further enhance model robustness and reduce overfitting. We have evaluated our approach on the UEC-Food100 dataset and observe substantial improvements in performance, particularly in terms of Top-1 and Top-5 classification accuracy.

The remainder of this paper is structured as follows. Section 2 reviews related work and techniques relevant to our study. Section 3 details our proposed training methodology, including progressive fine-tuning, hybrid label smoothing, and EMA distillation. Section 4 presents the experimental setup, results, and analysis. Finally, Section 5 summarizes our findings and outlines future directions.

2 Related Work

Fine-grained food classification has been an active research discipline of computer vision as this task serves for many real-world applications. Over the years, numerous researchers have made considerable efforts to handle this challenging issue through a variety of approaches, spanning from classical techniques to modern deep learning architectures.

In the early stages, several traditional approaches were employed to tackle this challenging task. Most of these methods relied on hand-crafted features combined with classical machine learning models such as support vector machines (SVMs) or decision trees. One of the earliest publications on fine-grained food classification is the work by Matsuda, Hoashi and Yanai (2012), in which they utilized features like SIFT, HOG, and Gabor filters in combination with a multiple kernel learning SVM to classify food items from images containing multiple dishes, achieving moderate performance on the UEC-Food100 dataset (2). Similarly in 2013, Kawano and Yanai proposed a mobile food recognition system that segments food regions using GrabCut, extracts color histograms and SURF-based features, and classifies them using a linear SVM with a fast χ^2 kernel (3). Bossard et al. (4) introduced the Food-101 dataset and proposed a discriminative part-mining method based on Random Forests and superpixels in 2014. These approaches demonstrated the feasibility of automated food recognition, but their reliance on manually crafted features limited their ability to generalize in the presence of large intra-class variation and inter-class similarity.

As the field progressed, more significant efforts was made to address limitations existed in conventional approach. The advent of deep learning models like convolutional neural networks (CNNs) began to dominate the food classification research due to their superior ability to learn discriminative features directly from raw image data. Of the first researchers using deep learning models in food recognition, Kagaya et al. (2014) introduced the implementation of fundamental CNN architecture in the recognition and detection of food images (5). Beside some traditional CNN models like AlexNet, GoogLeNet, we also witness the advent of numerous innovative models with higher efficiency, such as ResNet (6) and MobileNet (7). Later, these models are even improved better with both conventional and newer techniques to achieve the best possible performance. For instance, Liu et al. (2016) proposed fine-tune technique with Inception / GoogLeNet (8); Hassannejad et al. also utilized fine-tune method for InceptionV3 model to enhance the accuracy (9), and both these researchs conducted on three datasets Food-101, UEC-Food100, UEC-Food256, showing significant enhancement in statistics. In the same year, Martinel et al. showed even more groundbreaking results on UEC-Food100 with the combination of residual block and slice convolution (10).

3 Proposed Method

In this paper, we propose a deep learning-based approach to address the existing challenges in fine-grained food classification. Specifically, we adopt progressive fine-tuning on a pre-trained EfficientNetB0 model to leverage its learned representations and accelerate convergence. In addition, we incorporate label smoothing and Exponential Moving Average (EMA) distillation to improve generalization and enhance robustness against label noise and class imbalance.

3.1 EfficientNet-B0

EfficientNet-B0 is the baseline architecture of the EfficientNet family proposed by Tan and Le (2019), which introduces a principled approach to scaling convolutional neural networks. Instead of arbitrarily increasing network depth, width, or resolution, EfficientNet employs a compound scaling method that uniformly balances these three dimensions using a set of fixed scaling coefficients. This allows the model to achieve significantly better accuracy and efficiency trade-offs compared to previous architectures.

EfficientNet-B0 is built upon Mobile Inverted Bottleneck Convolution (MBConv) blocks, originally proposed in MobileNetV2, combined with squeeze-and-excitation (SE) optimization to adaptively recalibrate channel-wise feature responses. This design enables the network to capture both local and global patterns efficiently with fewer parameters and lower computational cost. The architectural details of EfficientNet-B0 can be seen in Figure 2.

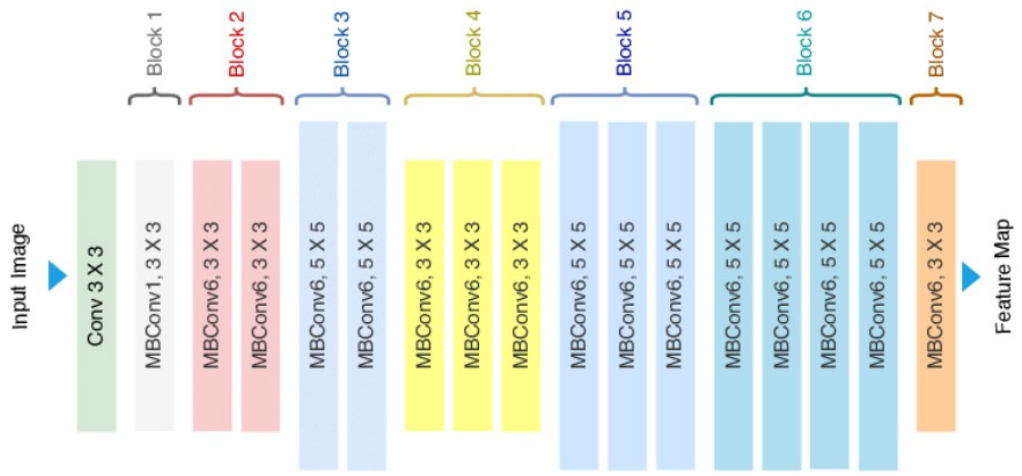


Figure 2: EfficientNet-B0 architecture.

Despite being the smallest model in the EfficientNet family, EfficientNet-B0 achieves remarkable performance on ImageNet with only 5.3 million parameters and 0.39 billion FLOPs, outperforming models like ResNet50 and InceptionV3 in both accuracy and efficiency. This makes it an ideal backbone for real-world applications, particularly in scenarios that require lightweight and deployable models, such as mobile food recognition apps.

In our work, we adopt EfficientNet-B0 as the base model for fine-grained food classification due to its strong balance between representational capacity and computational efficiency. While larger variants like EfficientNet-B7 could offer marginal gains in accuracy, they incur significantly higher resource demands, which are not desirable for real-time or resource-constrained applications. Through our proposed enhancements—progressive fine-tuning, label smoothing, and EMA distillation—we aim to extract the most from this compact yet powerful backbone for our target dataset, UEC-Food100.

3.2 Progressive Fine-Tuning

Progressive fine-tuning is a transfer learning approach that gradually adapts a pre-trained convolutional neural network (CNN) to a target task by incrementally unfreezing parts of the model during training. Instead of updating all weights at once—which can lead to instability or overfitting, especially when training on limited or fine-grained datasets—this method starts by fine-tuning only the final classification head, while keeping all pre-trained convolutional layers frozen. Over the course of training, deeper layers of the network are selectively unfrozen to allow more of the model to adapt to the target domain.

In our implementation, we apply progressive fine-tuning to a pre-trained EfficientNet-B0 model on the UEC-Food100 dataset. During the first five epochs, only the newly added classification head is trained, while the entire feature extractor remains frozen. This allows the model to begin adapting to the dataset without disturbing the general-purpose features learned from ImageNet. Starting from the sixth epoch, we unfreeze the last two convolutional blocks (features.6 and features.7) in addition to the classification head. The earlier layers remain frozen throughout to preserve foundational features such as edges and textures.

This strategy effectively reduces the risk of catastrophic forgetting by gradually exposing more layers to updates only when the model has already begun to specialize. It also mitigates overfitting by limiting the number of trainable parameters in the early stages of training.

To support stable optimization during this process, we combine progressive unfreezing with a warm-up and cosine annealing learning rate schedule. Although the learning rate is applied globally, the gradual increase in trainable depth, combined with the decreasing learning rate over time, helps mimic the behavior of layer-wise learning rate control.

Overall, progressive fine-tuning allows the model to retain the robustness of pre-trained features while gradually adapting to the specific characteristics of food images in the UEC-Food100 dataset.

3.3 Label Smoothing

Label smoothing is a regularization technique used to prevent a classification model from becoming overly confident in its predictions. In traditional supervised learning, ground-truth labels are represented as one-hot vectors, assigning a probability of 1 to the correct class and 0 to all others. While this strict labeling is effective in many scenarios, it can encourage the model to become overconfident, leading to poor generalization and increased vulnerability to noisy labels or minor variations in the input.

Label smoothing addresses this issue by softening the target distribution. Instead of assigning a hard 1 to the correct class, it assigns a slightly lower value (e.g., 0.85), and distributes the remaining probability mass uniformly across the other classes. This discourages the model from assigning full certainty to any single class, making the output distribution smoother and the learned representations more robust.

In our training setup, we incorporate label smoothing directly into the loss function by using PyTorch's CrossEntropyLoss with the label smoothing parameter set to 0.15. This means that during training, the target labels are smoothed such that 85% of the weight is placed on the correct class, and the remaining 15% is distributed evenly among the other 99 classes in the UEC-Food100 dataset. The modified objective encourages the model to generalize better by avoiding overfitting to noisy or overly specific features.

This is particularly important in our case, as the UEC-Food100 dataset contains many visually similar food items with subtle differences. By reducing the model's tendency to overfit to hard labels, label smoothing improves the robustness of the classifier and enhances its ability to distinguish between fine-grained categories. Combined with progressive fine-tuning and other regularization techniques like data augmentation, label smoothing contributes to stable and improved generalization performance during both training and evaluation.

3.4 EMA Distillation

EMA Distillation (Exponential Moving Average Distillation) is a knowledge distillation technique that leverages a slowly-updated teacher model to guide the training of a student model. Instead of using a separately pre-trained teacher, EMA distillation maintains a teacher model by continuously updating its weights as an exponential moving average of the student's weights throughout training. This approach encourages the student model to match a more stable and ensemble-like version of itself over time, leading to improved generalization and training stability. The EMA update rule can be expressed as:

$$\theta_{teacher} \leftarrow decay \cdot \theta_{teacher} + (1 - decay) \cdot \theta_{student} \tag{1}$$

where decay is a momentum-like factor (typically close to 1), ensuring that the teacher evolves more slowly and accumulates the "wisdom" of past student weights.

In our training pipeline, we implement EMA distillation by maintaining a frozen copy of the model (teacher model) which is initialized as a deep copy of the student model before training begins. During each training step, after the student model performs a gradient update, we update the teacher model using the EMA rule with a decay factor of 0.999. The teacher is then used to produce soft target logits for distillation. During training, the student is optimized using a combination of two losses:

- Hard target loss: A standard cross-entropy loss computed with the ground-truth labels.
- Soft target loss: A Kullback-Leibler divergence between the softened outputs of the student and the teacher, both scaled by a temperature hyperparameter T.

The final loss is a weighted sum of the hard and soft losses:

$$loss = \alpha \cdot \mathcal{L}_{KD} + (1 - \alpha) \cdot \mathcal{L}_{CE} \tag{2}$$

- $\mathcal{L}_{CE}$: Cross-entropy loss with label smoothing.
- $\mathcal{L}_{KD}$: KL divergence between soft predictions from the teacher and the student.
- α: A weighting factor that balances the importance of soft and hard targets.

This EMA-based distillation strategy stabilizes the learning process, especially during the early phases of training where the student model might be unstable. By encouraging the student to match a temporally smoothed version of itself, the model converges more reliably, gains better generaliztion and achieves improved accuracy, particularly on challenging datasets like UEC-Food100 that exhibit high inter-class similarity and intra-class variance.

4 Experiments

4.1 Dataset

We conduct experiments on the UEC-Food100 dataset, which contains 100 Japanese food categories with a total of over 14,000 images. Each class directory includes a bb_info.txt file that provides ground-truth bounding boxes. We crop each image to its bounding box region before applying any augmentations. The dataset is split into 80% training and 20% testing, ensuring both sufficient training data and a representative test set for evaluation.

4.2 Code Implementation

We implemented our method using PyTorch. The UEC-Food100 dataset was preprocessed by cropping each image to its annotated bounding box. We used EfficientNet-B0 pretrained on ImageNet as the backbone and replaced its classification head to fit 100 classes. To enhance generalization and stability, we applied extensive data augmentation, label smoothing, cosine annealing learning rate scheduling, and exponential moving average (EMA) teacher-student distillation. Progressive layer unfreezing was employed to gradually fine-tune deeper layers. The model was trained using a combination of cross-entropy and softened knowledge distillation loss.

4.3 Experimental Results

Table 1 presents a comprehensive comparison between our proposed method and a range of previous approaches on the UEC-Food100 dataset. As observed, our Hybrid EfficientNet-B0 achieves a Top-1 accuracy of 83.75% and a Top-5 accuracy of 96.1%, outperforming the baseline EfficientNet-B0 by a large margin of +11.5% Top-1 and +4.1% Top-5. Importantly, this improvement is attained while maintaining a low computational cost of only 0.4 GFLOPs, which is nearly identical to the original EfficientNet-B0 (0.39 GFLOPs). This highlights the effectiveness of our method in enhancing accuracy without compromising efficiency.

Table 1: Comparison of Accuracy and Computation Complexity between our proposed method and some notable previous methods

Method	Top-1 Accuracy	Top-5 Accuracy	FLOPs (G)
EfficientNet-B0 (baseline)	72.26%	92%	~ 0.39
AlexNet (11)	75.62%	92.43%	~ 0.72
Improved CNN-based GoogLeNet (8)	76.3%	94.6%	~ 1.5
Fine-tuned 54-layer Inception (9)	81.45%	97.27%	~ 11.5
NutriNet (12)	83.6%	99.2%	~ 1.6
Hybrid EfficientNet-B0 (Proposed)	**83.75%**	**96.1%**	$\sim$ **0.4**
ResNet200 (13)	86.25%	98.91%	~ 15
WISeR (10)	89.58%	99.23%	~ 5.5

Meanwhile, compared to other classic architectures such as AlexNet and Faster R-CNN, our method shows clear superiority both in accuracy and efficiency. AlexNet yields a Top-1 accuracy of 75.62% with 0.72 GFLOPs, while Faster R-CNN reaches 76.3% Top-1 but at an extremely high computational cost exceeding 100 GFLOPs. This reinforces the practicality of our lightweight hybrid design in real-world deployments.

In contrast, more complex architectures like Fine-tuned Inception (81.45%), NutriNet (83.6%), ResNet200 (86.25%), and WISeR (89.58%) achieve higher accuracy in some cases, but their significantly higher FLOPs (ranging from $\sim$ 1.6 to 15 GFLOPs) limit their applicability on resource-

constrained devices. For instance, WISeR achieves nearly 90% Top-1 accuracy but requires 5.5 GFLOPs—more than 13 times our proposed model's cost.

Our method demonstrates a compelling trade-off between recognition performance and computational demand. Through a combination of advanced data augmentation, label smoothing, cosine annealing learning rate scheduling, and EMA-based knowledge distillation, we effectively enhance generalization. Moreover, progressive fine-tuning of deeper layers contributes to the stability of training and improved convergence, especially in the early epochs.

5 Conclusion

In this work, we proposed a lightweight effective approach for food image classification on the UEC-Food100 dataset by enhancing original EfficientNet-B0 with a hybrid training strategy. Our method combines strong data augmentation, label smoothing, cosine annealing learning rate scheduling, and exponential moving average (EMA) teacher-student distillation. In addition, we introduced a progressive fine-tuning scheme to gradually unfreeze deeper layers of the network, which contributes to more stable optimization and improved generalization.

Experimental results show that our Hybrid EfficientNet-B0 significantly outperforms the original baseline and several previous methods in Top-1 and Top-5 accuracy, while maintaining an extremely low computational cost of just 0.4 GFLOPs. This demonstrates the model's strong balance between accuracy and efficiency, making it particularly suitable for deployment on devices with limited resources.

In future work, we plan to explore whether our hybrid training strategy can be effectively integrated into more heavyweight models such as EfficientNet-B7 or ResNet50, with the goal of surpassing the current state-of-the-art accuracy. Additionally, incorporating attention mechanisms or domain-specific priors—such as ingredient-level cues—may further enhance performance on fine-grained food recognition tasks.

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